L3IA Morocco at ArGuard 2026: Intent over Words - Uncovering Implicit Harm in Arabic LLM Prompts-

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Abstract

We present the L3IA Morocco team's system for shared Task B, designed for large language models and focused on detecting harmful prompts in Arabic. In this context, we define two subtasks: B1 and B2, which refer, respectively, to the binary safe/unsafe classification and the classification of harmfulness into 7 categories. Our system combines three fine-tuned Arabic encoders: AraBERTv2, CAMeLBERT-mix, and MARBERT, trained with class-weighted cross-entropy and combined by unweighted soft voting. It achieves a macro-F1 score of 0.946 for B1 and 0.902 for B2 during the development phase. In contrast, performance drops to 0.639 for B1 and 0.758 for B2 in the blind test. We demonstrate that this generalization divergence is due to a "unsafe" overprediction bias resulting from an unbalanced training corpus (80% unsafe) and not due to alignment of identifiers or dialects. Finally, we observed a negative result: in zero-shot and few-shot, AceGPT-7B, a generative Arabic LLM, performed poorly compared to fine-tuned encoders (0.38-0.42 on Darija).

1 Introduction

The increasingly widespread use of large language models (LLMs) among Arabic-speaking users highlights the critical importance of implementing safeguards that identify harmful prompts before they reach the model. Arabic presents specific challenges: irony, a wide variety of dialects, code-switching, cultural references, and, furthermore, a harmful intent that is often implied. An apparently simple prompt can mask a harmful intent, while sensitive language may indicate a real need for assistance: harmfulness lies in the intent, not in the surface-level vocabulary.

Furthermore, the actual safety evaluation of Arabic LLMs is conducted through the **ArGuard 2026** task (Alam et al., 2026). We are participating in Task B (textual), which is divided into two subtasks:

- **B1**: classifies a prompt safe or unsafe;
- **B2**: allows one of seven harmfulness categories to be assigned to the prompt (harm to others, self-harm, harassment, bullying, adult content, hate speech, fraud/deception).

Our contributions are threefold, as follows: (i) We propose a set of three complementary Arabic encoders that achieve 0.946 for B1 and 0.902 for B2, respectively, during development. **(ii)** We conduct an in-depth analysis by dialect and category, followed by a detailed evaluation of the generalization gap observed in the blind test, and identify its cause. **(iii)** We present a negative result demonstrating that, whether in zero-shot or few-shot, a generative LLM with 7 billion parameters is not competitive compared to fine-tuned encoders.

2 Related Work

The challenge of detecting harmful content hidden within Arabic prompts persists; various models have been implemented to optimize this detection.

Arabic encoders. AraBERT (Antoun et al., 2020) focuses on Modern Standard Arabic; recently developed, it has achieved remarkable performance on most of the Arabic NLP tasks tested. MARBERT (Abdul-Mageed et al., 2021), based on Arabic, relies on a deep bidirectional Transformer architecture. Furthermore, it was pre-trained on nearly one billion multi-dialectal tweets; CAMeLBERT (Inoue et al., 2021) combines Standard, Dialectal, and Classical Arabic and is also pre-trained on the largest dataset. In this context, we exploit their complementary strengths by combining their performance.

Safety and harmful content in Arabic. The detection of hate speech and offensive language has been the focus of numerous workshops, notably

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Open-Source Arabic Corpora and Processing Tools (OSACT) (Mubarak et al., 2020), which focused on the detection of offensive speech. Two subtasks were proposed: the first, Subtask A, concerns the identification of offensive speech, including insults and attacks, while the second, Subtask B, focuses on the detection of hate speech. The advent of LLMs has shifted attention toward the design of safeguards and benchmark datasets dedicated to Arabic safety, such as AraSafe (Mubarak et al., 2025), the first large-scale benchmark dataset to address the pressing need for evaluation resources that are representative of cultural and linguistic aspects. It comprises 12,000 human-created Arabic expressions, covering harmful and non-harmful content across multiple sectors. Our work takes place before the model is run, at the prompt filtering stage.

Arabic generative LLMs. Recent models include Jais, AceGPT, Llama-3, etc., with between 7 and 13 billion parameters (Mubarak et al., 2025). We evaluate AceGPT (Huang et al., 2024), an open-source Arabic model that covers the entire development process, including the pre-training phase, supervised fine-tuning, and reinforcement learning based on feedback from artificial intelligence acting as a judge; indeed, the integration of a dialect-sensitive model represents a promising avenue for us.

Connection to sentiment analysis. Harmfulness detection and sentiment analysis are related. Indeed, both involve the classification of subjectivity and aim to model meaning that goes beyond surface-level lexical indicators, including irony, subjectivity, negation, and dependence on context and dialect. The attention and ensemble architectures designed for sentiment analysis (Saidi et al., 2025a, Saidi et al., 2025b) are therefore directly applicable; we adapt this methodological approach to LLM safety, highlighting (in Section 7) a crucial distinction between these two tasks regarding the abusive use of the model's confidence.

3 Task and Data

3.1 Corpus Composition

Task B is a purely textual (unimodal) task : all inputs are Arabic text prompts directed at an LLM, with no image or audio modality. The corpus contains approximately 25,000 annotated prompts: 16,833 for training (after removing two missing labels) and 2,405 for development with reference labels. The final stage is based on two distinct blind test sets: 7,809 prompts for Task B1 and 5,979 for Task B2 (Zaghouani et al., 2026,Mubarak et al., 2025). Each example includes the following elements: an identifier, the text, the dialect, a binary label, and a category label. The binary label set exhibits a marked imbalance: 80% unsafe cases versus 20% safe cases, an imbalance that, as we will demonstrate, influences the model's performance on generalization. The harmful content categories range from 4,592 adult content examples to 943 examples of hate speech.

3.2 The intent is orthogonal to the subject

Our approach is based on a key characteristic of the corpus: the partial independence between the harm category and the safety label (Appendix A). Self-harm has 985 safe examples compared to 1,737 unsafe ones; harm to others has 619 safe examples compared to 2,421 unsafe ones; bullying, fraud, and hate speech are all classified as unsafe. In categories related to intent, the same vocabulary can be safe (victim seeking assistance) or unsafe (aggressor): harmfulness is determined at the level of intent.

4 Proposed System

The overall architecture is illustrated in appendix B.

4.1 Preprocessing and Base Models

Given that prompts are generally concise, with a median of 10 words, we limit their maximum length to 96 subwords and apply minimal cleaning, thereby preserving dialectal spelling. We fine-tune three encoders from the BERT family, selected for the variety of their pre-training: MARBERT (dialectal), CAMeLBERT-mix (mixed), and AraBERTv2 (standard). Regarding the choice of method, the seven categories across all prompts are evaluated using the official B2 score; training B2 on the entire corpus, given that training on unsafe prompts alone significantly degrades the result (0.81 vs. 0.90).

4.2 Weighting, assembly and LLM

We use weighted cross-entropy (balanced heuristic): in the case of B1, the weights are approximately 2.48 (safe) versus 0.63 (unsafe), to prevent the minority class from being underestimated by the macro-F1 score. The three models are combined by unweighted soft voting: we average their softmax probability distributions with equal weight

(1/3 each) and take the argmax. We deliberately use equal weights rather than tuned weights, since the three encoders achieve comparable individual performance (Appendix C). We also considered AceGPT-7B-chat quantized to 4 bits as an alternative evaluator, as described in Section 6.4.

5 Experimental Protocol

Training on an NVIDIA T4 GPU. Official metric: macro-F1. Hyperparameters: batch size 32, rate 2×10^{-5}, warm-up 0.1, decay 0.01, mixed precision, 3 epochs (B1) and 4 (B2), early stopping based on the development macro-F1. Our local scores match the official scorer to the closest fourth decimal, ensuring the accuracy of our measurements.

Test configuration. The submitted predictions were produced by the three-encoder ensemble, each model trained on the training split only (not train+dev). At inference, the three softmax distributions are averaged with equal weight and the argmax is predicted; for B1 this corresponds to a standard 0.5 decision threshold on the unsafe probability. No per-class threshold tuning or prior correction was applied to the submitted run. The training, development, and test splits are provided as disjoint sets by the organizers, with no overlapping prompts; no test data was used at any stage of training or model selection, ruling out data leakage.

6 Results

6.1 Overall Performance (Development)

The progression from the single model to the ensemble is illustrated in Appendix C and D . The three-model ensemble consistently outperforms the best single model: +0.25 points on B1 and +1.6 points on B2. The significant improvement observed for B2 is due to the greater margin of complementarity provided by a seven-class task. While the ensemble gain over the best single model is modest on B1, it is more substantial on B2(+1.6 points) and, importantly, improves robustness across dialects and reduces variance compared to any individual encoder.

6.2 Analysis by Dialect

The performance is detailed by variety in Appendix E and F. Maghreb variants such as Algerian Darija are the most complex, with a difference of up to 14 points compared to the Egyptian dialect, highlighting their underrepresentation in the pretraining corpora, which raises fairness concerns for the safety of Arabic LLMs.

6.3 Analysis by Category and Confusion

Errors occur primarily in categories dependent on intent: "harm to others" acts as an attractor, including confusion related to harassment, bullying, and self-harm, whereas categories with explicit lexical markers (adult content: F1 = 0.990; hate speech: 0.933) are virtually error-free. This contrast supports our argument: classifications based on vocabulary are simple, whereas those based on intent are complex.The confusion matrix for B2 is presented in Appendix G.

6.4 Negative result: the generative LLM

Guided by the idea that a generative model would be more effective at capturing the underlying intent, we tested AceGPT-7B on darija, the most challenging dialect, for Task B1. Appendix H summarizes the failure. The model labels nearly all prompts as "unsafe" (overcautious, macro-F1 = 0.424) in zero-shot; it tends to overcorrect toward "safe" (145 safe predictions versus 62 actual ones, macro-F1 = 0.385) in few-shot. In the few-shot setting, we used six demonstrations (three safe, three unsafe), all drawn from the training split and written in Moroccan Darija to match the target dialect; the zero-shot setting used none. In both cases, generation was constrained to a single safe/unsafe label. Furthermore, the encoders' confidence proved to be uninformative (incorrect predictions at approximately 0.90 confidence), eliminating any preference for targeting based on confidence.

6.5 Generalization Gap from Development to Test

On the blind test sets, performance drops significantly: 0.639 for task B1 and 0.758 for B2, compared to 0.946 and 0.902 during development (Appendix I). Such a gap (30 points on B1) requires investigation; we systematically eliminate the likely causes.

Analysis of the alignment problem. The test identifiers range continuously from 1 to 7,809. They are correctly paired and unique. The submission format is identical to the validated format from the development phase.

Dialect effect analysis. At first glance, the test contains more Standard Arabic (68%, compared to 48% during training) and fewer Maghrebi dialects (Appendix J, left side). Consequently, from this perspective, the test is more optimal for our model. Dialect cannot justify the decline.

Analysis of "unsafe" overprediction bias. In Test B1, the model predicted 6,252 prompts classified as unsafe versus 1,557 classified as safe (Appendix J, right), which corresponds exactly to the 80/20 ratio of the training corpus. Qualitative analysis corroborates this finding: apparently harmless prompts ("thank you for this information," "who is Fatima Zahra," "how many letters are in the word Ramses") are categorized as unsafe with a high level of confidence. This type of factual question dominates the test, with a total of 1,621 short prompts ($\leq$ 6 words) classified as unsafe.

Study of the learning process behind non-calibration biases. Changing the decision threshold from 0.30 to 0.60 does not alter the macro-F1 score during development ($\approx$ 0.946): no threshold adjustment eliminates the bias. Training the model on a corpus containing 80% unsafe prompts, which generally represent short, direct malicious requests, led the model to learn a surface heuristic that classifies short prompts directly as unsafe. This heuristic works well on a development dataset with a similar distribution but fails in tests containing many insignificant factual questions.

6.6 Mitigating the unsafe over-prediction bias

Building on the diagnosis above, we investigate whether the development-to-test collapse can be corrected post-hoc, using the released gold test labels. The B1 test set turns out to be nearly balanced (51% safe/49% unsafe), in sharp contrast with the 80/20 training distribution, which confirms that the gap is driven by a label-distribution shift rather than by a modelling failure. Appendix K reports two corrections applied to the submitted ensemble. A prior correction that reweights the predicted probabilities by the inverse training prior, using no test information, recovers +8.4 macro-F1 points ($0.639 \rightarrow 0.723$). An oracle threshold tuned on the test recovers +14.2 points ($0.639 \rightarrow 0.781$), which we report as an upper bound. Both corrections primarily restore the safe class, whose recall rises from 0.37 to 0.67. Most of the gap is therefore a recoverable calibration problem tied to the training imbalance, rather than an intrinsic limitation of the encoder ensemble. This confirms that a distribution-aware correction is the natural remedy for the diagnosed bias.

7 Discussion

Our results point toward a unified interpretation. Errors occur in the categories dependent on intent and in the Maghrebi dialects during the development phase, where surface indices are unreliable; however, in the testing phase, the model's dependence on these surface indices, particularly length and vocabulary, causes the model to fail: the model reacts to the form rather than the meaning when faced with a short, polite prompt that carries no harmful intent. This observation supports our main idea: understanding intent based on words cannot rely on superficial correlations. In this sense, the title's emphasis on "intent over words" is prescriptive rather than descriptive : our analysis precisely demonstrates that surface-level models fail to capture intent, motivating intent-aware approaches rather than claiming that our system already achieves this. The corpus imbalance, which is beneficial for optimizing macro-F1 during development, leads to an unfavorable bias during generalization. Furthermore, the generative LLM, although capable of reasoning correctly in isolated cases, fails at scale due to calibration instability: the fluidity of ad hoc reasoning does not guarantee reliable classification. Intent detection relies on carefully calibrated supervised learning, not on the model's scale. Based on previous studies on sentiment analysis, an approach based on model confidence, which activates only heavier agents for uncertain cases, has proven effective (Saidi et al., 2025b). This task presents an illuminating contrast: the encoders' confidence does not provide any information, as incorrect predictions retain a high degree of confidence ($\approx$ 0.90). This divergence indicates that the reliability of calibration is intrinsically linked to the nature of the objective when dealing with two tasks that share the same architecture: affective polarity, which is more directly associated with lexical indices, lends itself more readily to confidence evaluation than malicious intent, which is more implicit.

8 Conclusion and perspectives

We introduced a set of three Arabic encoders that achieved a performance score of 0.946 for B1 and 0.902 for B2 during the development phase. Beyond these evaluations, our major contribution lies in the analysis: the detection, in the blind test,

of a generalization divergence caused by an excess of "unsafe" predictions resulting from training imbalance, which we prove is neither calibrated nor dialectal. We also note the failure of a generative LLM model in zero-shot/few-shot scenarios. These findings point to concrete avenues for improvement: (i) a targeted increase in safe, innocuous prompts, particularly factual and cultural questions, to reduce bias, an intuition confirmed by our mitigation experiment (section 6.6), where a simple prior correction already recovers a large part of the gap; (ii) judicious corpus distribution to reduce reliance on superficial features; (iii) the application of the structures designed for sentiment analysis, notably genetic optimization of attention (Saidi et al., 2025a) and adaptive combination of agents with inter-agent communication (Saidi et al., 2025b), to the detection of harmful content; (iv) the incorporation of an Arabic LLM adapted to dialects as a calibrated judge; and (v) expansion toward a multimodal approach (Task A).

Limitations

This work is based on a single dataset and examines the generative LLM only in relation to Darija for Task B1. Some annotations raise ambiguities regarding the distinction between safe and unsafe content, particularly with regard to self-harm, which limits the achievable level of performance.

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A Safe/Unsafe Distribution by category (Training)

Category	Safe	Unsafe	Intent
Bullying	0	1 518	No
Fraud/Deception	0	1 764	No
Hate Speech	0	943	No
Adult Content	942	3 650	Yes
Harassment	850	1 404	Yes
Harm to Others	619	2 421	Yes
Self-Harm	985	1 737	Yes

Table 1: Safe/Unsafe Distribution by category (Training). Four out of seven categories are determined by the intent.

B Overall architecture of our system

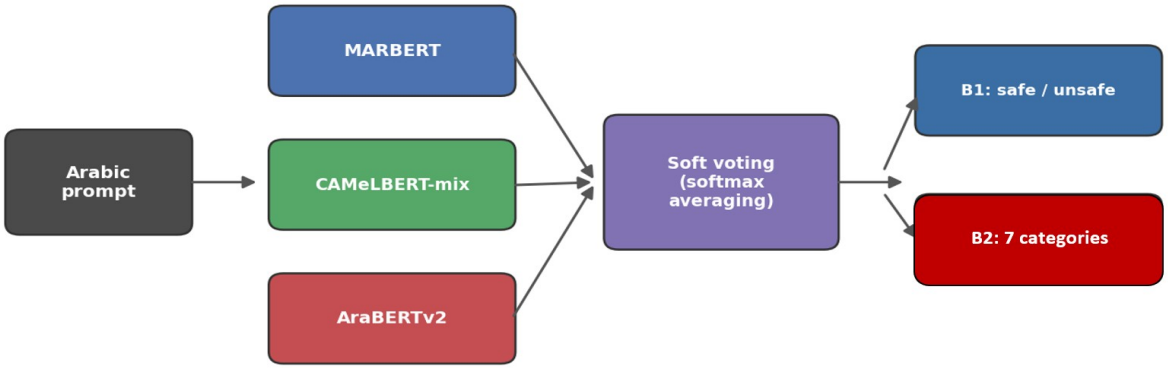


Figure 1: Architectural structure showing the three refined encoders that generate probability distributions, feeding the two subtasks through a combination based on soft voting.

C Macro-F1 in development

System	B1	B2
MARBERT	0,9389	0,8869
CAMeLBERT	0,9439	0,8862
AraBERTv2	0,9355	0,8582
Ens (2 mod.)	0,9451	0,8986
Ens (3 mod.)	0,9464	0,9024

Table 2: Macro-F1 in development. The last row refers to the submitted system.

D Evolution of the macro-F1

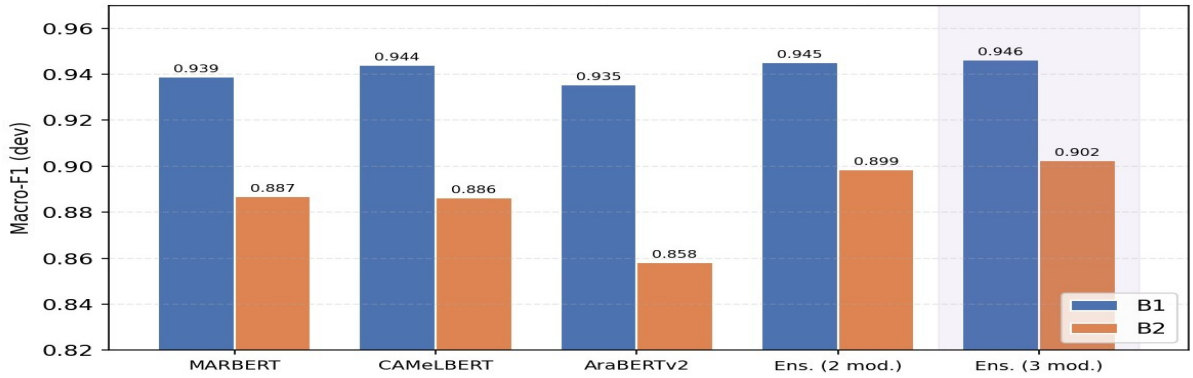


Figure 2: Evolution of the macro-F1 by system and subtask.

E Overall accuracy by dialect (development)

Dialect	B1	B2
Egyptian	0,991	0,986
Standard (MSA)	0,972	0,912
Syrian	0,972	0,914
Palestinian	0,972	0,881
Algerian	0,952	0,819
Darija	0,898	0,843

Table 3: Overall accuracy by dialect (development).

F Performance by dialect

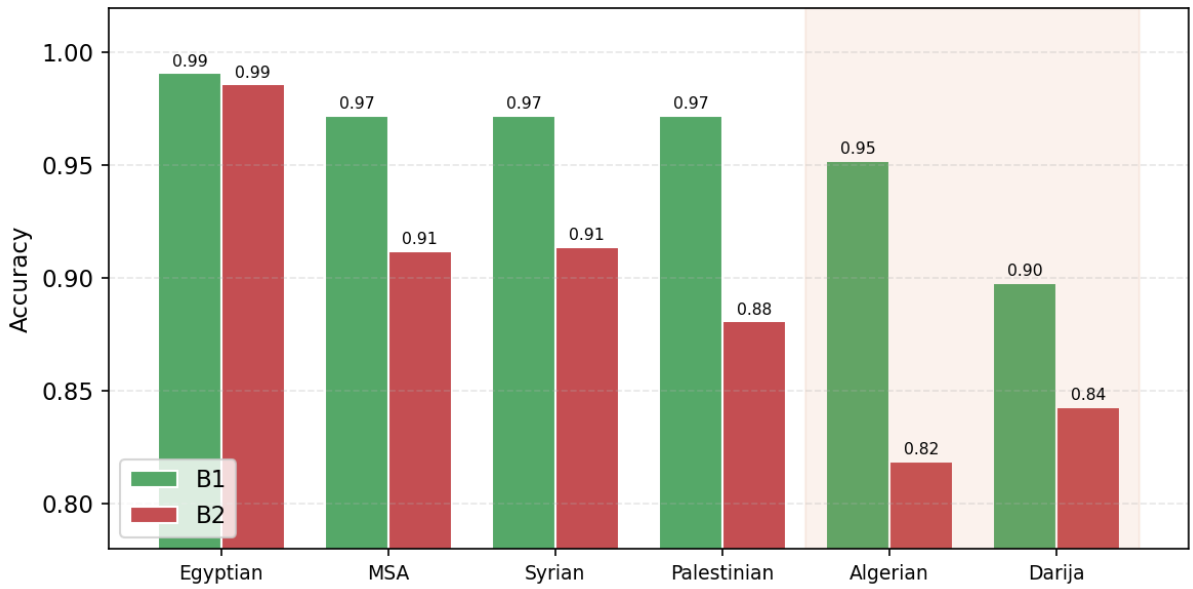


Figure 3: Performance by dialect, highlighting the Maghreb deficiency.

G Normalized confusion matrix (B2)

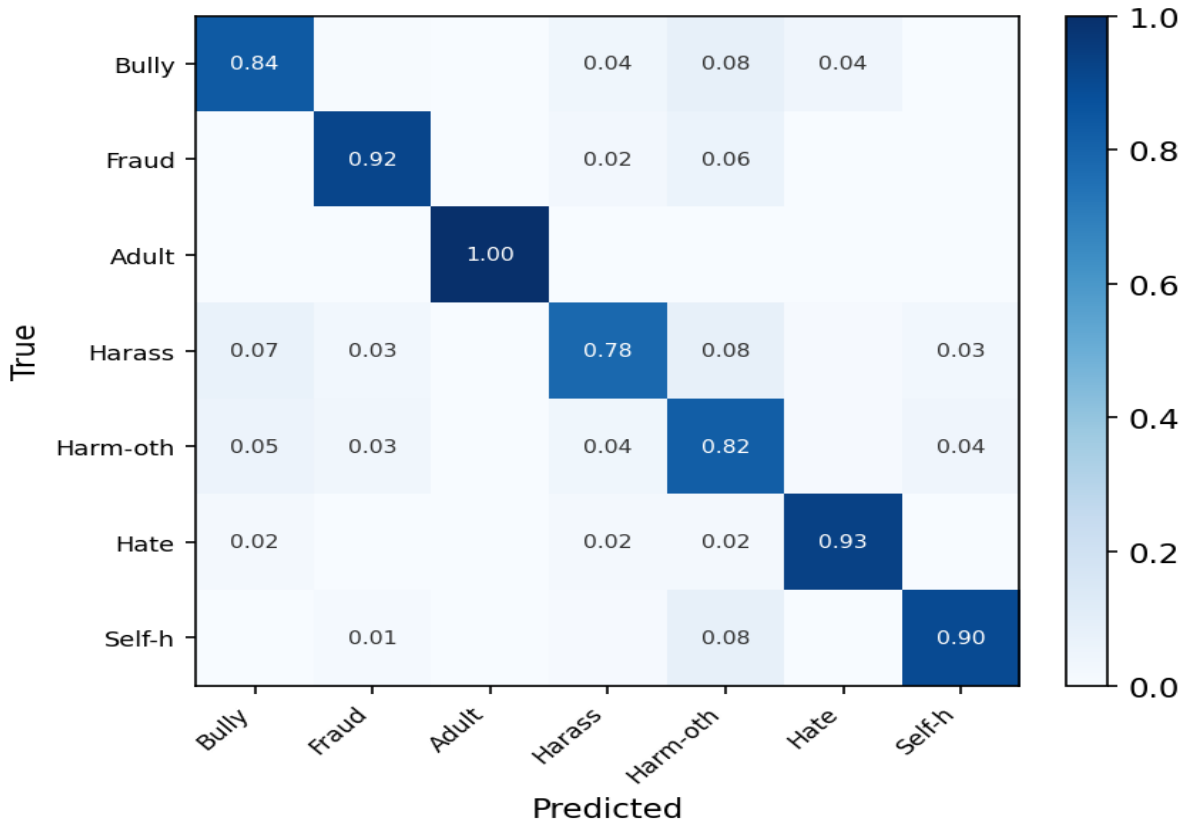


Figure 4: Normalized confusion matrix (B2). "Harm to others" attracts confusion from neighboring intent categories.

H Macro-F1 on Darija

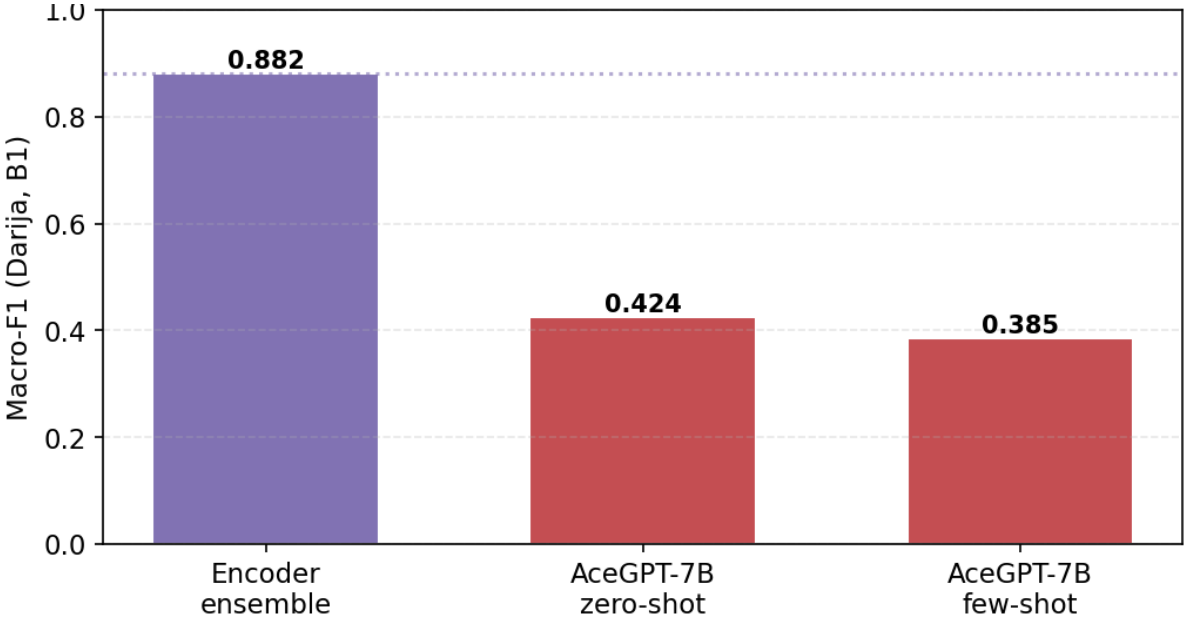


Figure 5: Macro-F1 on Darija. The generative LLM underperforms compared to the set of encoders.

I Generalization gap

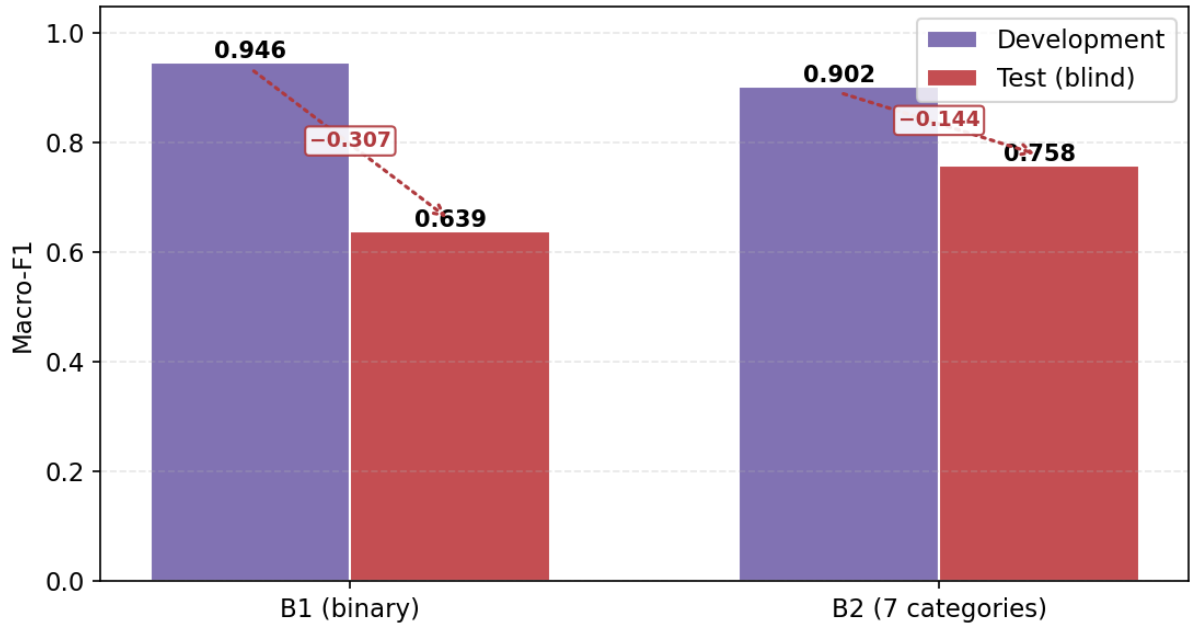


Figure 6: Generalization gap.

J Generalization Gap from Development to Test

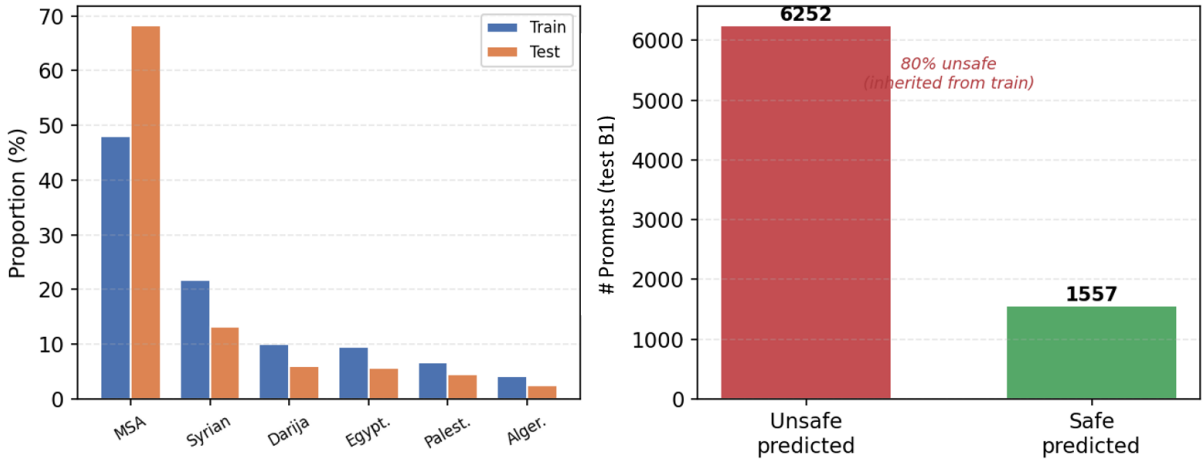


Figure 7: On the left, the test set includes a larger number of MSAs and a smaller number of Maghrebi dialects than the training set. On the right, the model replicates the 80/20 imbalance in the test data, as learned during training.

K Post-hoc bias mitigation on the B1 test set (gold labels)

Method	Macro-F1	Acc	F1-safe	F1-unsafe	Rec-safe
Submitted (thr. 0,5)	0,639	0,670	0,533	0,745	0,371
Prior correction	0,723	0,734	0,669	0,778	0,530
Oracle thr. (0,95)	0,781	0,783	0,758	0,803	0,671

Table 4: Post-hoc bias mitigation on the B1 test set (gold labels). Macro-F1: macro-averaged F1; F1-safe/F1-unsafe: F1 for the safe/unsafe class; Rec-safe: safe-class recall. The prior correction uses no test information.